

SMART CLASSROOM IOT NETWORK FOR MONITORING STUDENT MENTAL WELL-BEING AND BEHAVIOUR

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Specializing in Computer Systems and Network Engineering

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DECLARATION

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ABSTRACT

Smart Classroom IoT Network for Monitoring Student Mental Well-being and Behaviour

This group dissertation presents an integrated Smart Classroom IoT Network platform developed to address the technological, security, privacy, and environmental challenges of contemporary educational settings. The platform combines four interdependent components: adaptive IoT environmental and attendance monitoring; intelligent Software-Defined Networking (SDN) traffic classification and Quality of Service (QoS) enforcement; decentralized Zero Trust Network Access (DZTNA) with AI-based anomaly detection; and a privacy-preserving classroom engagement monitoring system using edge AI with differential privacy.

The IoT environmental monitoring component integrates an ESP32 microcontroller with DHT22, MQ-7, MFRC522, and sound sensor modules, recording classroom temperatures averaging 29.6°C and humidity averaging 72.5% RH across ten sessions, with 94.3% RFID attendance accuracy. A Q-Learning reinforcement learning middleware achieves 72.8% alert decision accuracy, surpassing static threshold baselines by 11.4 percentage points. The SDN traffic classification component employs Logistic Regression on flow-level and contextual features to distinguish academic from non-academic traffic in real time, enforcing DSCP prioritization and meter-based rate limiting without payload inspection.

The DZTNA security component achieves 87 ms median authentication latency, zero false authorizations across 500 authentication tests, and a macro-averaged F1-score of 0.922 across seven attack categories using TensorFlow Lite edge inference, all while maintaining offline resilience through local blockchain replication. The privacy-preserving engagement monitoring component achieves 84% engagement classification accuracy with a $358,293\times$ bandwidth reduction compared to video streaming, applying $\epsilon = 1.3$ Laplace differential privacy and 100% frame deletion to ensure mathematically rigorous student data protection.

Together, these four components form a cohesive, affordable, and privacy-respecting smart classroom infrastructure suitable for resource-constrained educational institutions in developing regions. The platform demonstrates that environmental quality monitoring, secure network management, Zero Trust access control, and privacy-preserving engagement analytics can be delivered as an integrated system without dependency on expensive cloud infrastructure or commercial security subscriptions.

Keywords: Smart Classroom, Internet of Things, Software-Defined Networking, Zero Trust Security, Differential Privacy, Reinforcement Learning, Edge AI, Engagement Monitoring, Blockchain Authentication, Q-Learning

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Special thanks are extended to the thirty student volunteers at SLIIT who participated in classroom monitoring sessions, contributing the real-world data that underpins the empirical results presented in this dissertation. Their patience and cooperation during the ten monitoring sessions were indispensable.

We acknowledge the open-source communities behind Hyperledger Fabric, Ryu SDN Framework, TensorFlow Lite, DeepFace, OpenCV, IBM diffprivlib, the MERN stack, and the Arduino IDE ecosystem. The availability of these tools has democratized access to enterprise-grade technologies, making research of this scope achievable by a group of undergraduate students working in resource-constrained environments.

Finally, we thank our families for their unwavering encouragement and understanding throughout the intensive research and writing phases of this project. Their support has been a constant source of motivation throughout our academic journey.

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LIST OF ABBREVIATIONS

Abbreviation	Description
ADC	Analog-to-Digital Converter
AI	Artificial Intelligence
API	Application Programming Interface
ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineers
BLE	Bluetooth Low Energy
CDN	Content Delivery Network
CO	Carbon Monoxide
CRUD	Create, Read, Update, Delete
DDoS	Distributed Denial of Service
DP	Differential Privacy
DSCP	Differentiated Services Code Point
DSR	Design Science Research
DZTNA	Decentralized Zero Trust Network Access
ECC	Elliptic Curve Cryptography
ESP32	Espressif Systems 32-bit Microcontroller
FER	Facial Expression Recognition
FL	Federated Learning
GDPR	General Data Protection Regulation
GPIO	General Purpose Input/Output
HTTP	Hypertext Transfer Protocol
IoT	Internet of Things
JSON	JavaScript Object Notation
MAC	Media Access Control
MDP	Markov Decision Process
MERN	MongoDB, Express.js, React, Node.js
MFRC522	Mifare RC522 RFID Reader Module

ML	Machine Learning
MQ-7	Metal Oxide Semiconductor CO Sensor Module
MQTT	Message Queuing Telemetry Transport
NIST	National Institute of Standards and Technology
NoSQL	Non-Structured Query Language (Database)
OpenFlow	Open Flow Protocol
OVS	Open vSwitch
ppm	Parts Per Million
QoS	Quality of Service
REST	Representational State Transfer
RFID	Radio Frequency Identification
RH	Relative Humidity
RL	Reinforcement Learning
SaaS	Software as a Service
SDN	Software-Defined Networking
SHA	Secure Hash Algorithm
SLIIT	Sri Lanka Institute of Information Technology
SNI	Server Name Indication
SPI	Serial Peripheral Interface
TLS	Transport Layer Security
TWA	Time-Weighted Average
UI	User Interface
ZT	Zero Trust
ZTNA	Zero Trust Network Access

CHAPTER 1: INTRODUCTION

1.1 Background and Literature Survey

1.1.1 IoT in Educational Environments

The Internet of Things (IoT) has rapidly evolved from a theoretical concept in machine-to-machine communication into a pervasive technological paradigm embedded across industries including healthcare, manufacturing, smart cities, and, increasingly, education. At its core, IoT refers to the interconnected network of physical devices embedded with sensors, actuators, and wireless connectivity that enables continuous data collection and automated responses without direct human intervention [1]. In educational contexts, this paradigm has given rise to the concept of the smart classroom — an environment in which digital infrastructure extends beyond audio-visual equipment to encompass real-time monitoring of physical, behavioural, and network parameters that directly influence student wellbeing and academic performance.

Surveys of IoT deployment in higher education institutions across developing regions indicate that over 65% of newly constructed campus facilities incorporate IoT-enabled monitoring infrastructure as of 2024, with projections suggesting near-universal adoption by 2030 [2]. The ESP32 microcontroller, manufactured by Espressif Systems, has emerged as the platform of choice for educational IoT prototyping due to its dual-core 240 MHz processor, integrated Wi-Fi and Bluetooth, 34 programmable GPIO pins, and unit cost under LKR 1,500 — a combination that makes it uniquely suitable for resource-constrained institutional deployments [3]. However, the rapid expansion of IoT device fleets in educational settings has outpaced the development of commensurate security, network management, and privacy frameworks, creating critical technical gaps that this research addresses.

1.1.2 Environmental Monitoring and Smart Classrooms

A substantial body of research confirms that indoor environmental quality directly impacts student cognitive performance, mood, and attendance behaviour. Wyon and

Wargocki [4] established through controlled experiments that task performance by school students declined by approximately 10% when temperatures exceeded 30°C — a threshold regularly exceeded in Sri Lankan classrooms without active cooling. Thermal comfort, relative humidity, carbon monoxide levels, and ambient noise are the primary environmental parameters identified in smart classroom studies as having measurable impact on student concentration and academic outcomes.

Prior IoT environmental monitoring implementations, such as the ESP8266-based indoor monitoring system by Marques et al. [5], demonstrated the technical feasibility of real-time sensor data collection in institutional buildings. However, these systems employed static, manually configured alert thresholds that do not adapt to observed environmental patterns, resulting in high false-positive alert rates and alert fatigue among administrators. The application of reinforcement learning to adaptive threshold management represents a significant advancement over static approaches, enabling IoT middleware to learn optimal alert policies from historical environmental data [6].

Traditional paper-based attendance systems are labour-intensive, error-prone, and susceptible to proxy fraud — a well-documented concern in higher education. RFID-based attendance systems using the MFRC522 module have been validated in academic research at detection accuracies exceeding 90%, offering a low-cost and reliable alternative that integrates naturally with IoT environmental monitoring nodes [7]. The integration of attendance management and environmental sensing within a single IoT deployment represents a novel architectural contribution not present in existing literature.

1.1.3 Software-Defined Networking for Educational Networks

The rapid advancement of information and communication technologies in education has transformed traditional classrooms into smart learning environments where students rely heavily on internet-connected devices for academic resources. While this transformation enhances learning experiences, it simultaneously introduces critical network management challenges, most notably the misuse of available bandwidth for non-academic purposes including social media browsing, video streaming, and online

gaming. Studies indicate that such activities consume a significant proportion of available network bandwidth, reducing performance for critical educational applications such as video conferencing and learning management platforms [8].

Software-Defined Networking (SDN) has emerged as a compelling paradigm for addressing these challenges by separating the network control plane from the data plane, enabling centralized and programmable network management [9]. The Ryu SDN controller, an open-source Python-based framework, provides a flexible platform for implementing custom network control applications through its event-driven architecture and OpenFlow 1.3 protocol support. Traditional traffic classification methods — including port-based identification and deep packet inspection (DPI) — are no longer effective in modern encrypted network environments. DPI is computationally expensive and raises significant privacy concerns, while port-based methods fail due to dynamic port usage and the widespread adoption of HTTPS [10].

Machine learning-based approaches offer a privacy-preserving alternative to DPI by operating on flow-level statistical features such as packet size distributions, inter-arrival times, flow duration, and byte counts, which are observable even for encrypted traffic. Logistic Regression is particularly attractive for SDN controller integration due to its computational efficiency, interpretability, and suitability for real-time binary classification tasks [11]. Combining flow-level statistical features with contextual signals derived from DNS queries and TLS Server Name Indication (SNI) significantly improves classification accuracy without requiring payload decryption, enabling differentiation between academic and non-academic traffic in a privacy-compliant manner.

Quality of Service enforcement via DSCP marking and OpenFlow meter tables enables dynamic prioritization of academic traffic and rate limiting of non-academic flows. This approach ensures that critical educational applications receive consistent network performance even during periods of peak bandwidth demand, directly supporting improved learning outcomes in technology-dependent classroom environments [12].

1.1.4 Zero Trust Security and Blockchain Authentication in IoT

The dominant paradigm for IoT network security in educational settings relies on centralized authentication servers that validate device identities and issue access tokens. While conceptually straightforward, this architecture introduces critical vulnerabilities that are particularly acute in educational IoT environments. The most significant structural weakness is the single point of failure it creates: when the central authentication server becomes unavailable — whether through network partition, hardware failure, or deliberate denial-of-service attack — all dependent devices lose authentication capability, rendering the entire smart classroom network inoperative. Empirical studies attribute 43% of educational institution network downtime to authentication service failures [13]. Cloud-dependent authentication also introduces 50–200 ms of additional latency per access decision and creates privacy risks for sensitive student data transmitted to external cloud providers.

The Zero Trust security model, first articulated by Forrester Research in 2010 and formalized by NIST in Special Publication 800-207 (2020), fundamentally rejects the perimeter-based trust model in favour of continuous verification. The core principle — 'never trust, always verify' — mandates that no network entity receives implicit trust based on its network location, prior authentication, or device type. Every access request must be explicitly authenticated, authorized, and continuously validated throughout its session [14]. Implementation of Zero Trust in IoT environments presents particular challenges not fully addressed by NIST SP 800-207, which was designed primarily for enterprise IT contexts with adequate computational resources.

Blockchain technology offers a compelling solution to the single-point-of-failure vulnerability of centralized authentication by distributing the device identity registry across multiple nodes in a tamper-proof ledger. Hyperledger Fabric, an enterprise-grade permissioned blockchain developed under the Linux Foundation's Hyperledger project, supports pluggable consensus mechanisms and modular architecture optimized for the throughput requirements of IoT authentication workloads [15]. Previous research has demonstrated that distributed ledger approaches can achieve

authentication latency comparable to centralized systems while eliminating single-point-of-failure vulnerabilities [16].

AI-based anomaly detection addresses the fundamental limitation of authentication-only security: authorized devices may exhibit malicious behaviour after successfully authenticating. TensorFlow Lite, Google's inference-optimized deployment framework, enables deployment of trained neural network models on resource-constrained edge devices at 15–30 ms inference latency — acceptable for real-time network security monitoring without requiring cloud inference infrastructure [17]. The combination of blockchain-based identity management, SDN-enforced access control, and AI-driven behavioural anomaly detection creates a layered defense-in-depth security architecture suitable for educational IoT deployments.

1.1.5 Edge AI and Privacy-Preserving Educational Monitoring

The intersection of artificial intelligence, edge computing, and educational technology has emerged as one of the most consequential research frontiers of the mid-2020s. Educational monitoring in its earliest computational forms comprised simple interaction logs and time-on-task metrics. The evolution of broadband internet facilitated the emergence of richer monitoring paradigms, most notably the integration of video conferencing into formal educational delivery. However, standard 720p video streaming at 24 frames per second consumes approximately 2–5 Mbps per student, requiring 60–150 Mbps of upstream bandwidth for a classroom of 30 students — well beyond the capacity of institutional internet connections in resource-constrained environments [18]. This fundamental constraint has historically precluded video-based engagement monitoring from institutions lacking premium infrastructure.

Facial expression recognition (FER) as a proxy for student engagement state has benefited enormously from deep learning advances. Convolutional neural networks trained on datasets such as FER2013 and AffectNet have progressively improved recognition accuracy from below 70% to above 85% [19]. The DeepFace library provides a particularly accessible implementation wrapping multiple pre-trained models within a unified Python interface [20]. Multi-modal fusion of emotion

recognition, head movement analysis, and face visibility tracking consistently outperforms unimodal approaches in classroom engagement detection tasks [21].

Differential privacy, introduced by Dwork and Roth [22], provides a mathematically rigorous framework for publishing statistical analyses of sensitive datasets while providing provable guarantees against individual inference. The formal privacy budget parameter ϵ controls the privacy-utility tradeoff: smaller values provide stronger protection at the cost of greater statistical noise. NIST guidelines classify ϵ values in the range 1–2 as providing strong differential privacy protection. The Laplace mechanism achieves ϵ -differential privacy for real-valued queries by adding noise drawn from a Laplace distribution scaled by the query's sensitivity — making it optimal for engagement proportion statistics computed over aggregation windows [22]. Edge processing architectures that apply differential privacy before any data crosses the network boundary simultaneously address bandwidth constraints and regulatory compliance requirements under GDPR and FERPA.

1.2 Research Gap

The literature survey reveals four interconnected and substantiated gaps that motivate this research. Together, they define the problem space for the integrated Smart Classroom IoT Network platform developed in this project.

First, no existing low-cost classroom IoT deployment integrates multi-parameter environmental sensing with RFID attendance monitoring within a single IoT node, sharing a unified data pipeline, dashboard, and adaptive middleware. Existing systems treat environmental monitoring and attendance management as isolated, independently deployed systems, forfeiting the analytical potential of cross-parameter correlation. Furthermore, alert systems in educational IoT deployments universally employ static, manually configured thresholds that do not adapt to observed environmental patterns, producing chronic alert fatigue.

Second, existing SDN-based traffic management systems for educational environments have not been comprehensively evaluated under realistic conditions

combining encrypted traffic, CDN-based routing, and concurrent academic and non-academic usage. The majority of machine learning-based classification approaches are evaluated using offline datasets rather than live SDN controller deployments, and few integrate classification directly with dynamic QoS enforcement through DSCP marking and meter-based rate limiting.

Third, no existing solution designed for resource-constrained educational IoT deployments simultaneously combines all three critical security layers: blockchain-based decentralized identity management, SDN-enforced Zero Trust access control, and AI-driven behavioural anomaly detection. Commercial solutions such as Azure IoT Hub Defender provide partial Zero Trust enforcement but require cloud connectivity and per-device subscription costs prohibitive for developing-region institutions. The specific integration of the Ryu-Blockchain middleware architecture — bridging OpenFlow SDN enforcement with blockchain-validated Zero Trust access control in the OpenFlow Packet-In processing loop — is a novel architectural contribution absent from prior literature.

Fourth, no existing educational monitoring system simultaneously achieves mathematical differential privacy guarantees ($\epsilon \leq 2$), extreme bandwidth efficiency (< 1 Kbps per student), and acceptable engagement classification accuracy ($> 80\%$) for practical classroom deployment in resource-constrained environments. Commercial monitoring platforms such as Microsoft Teams Analytics and Google Classroom transmit raw video data without privacy protection, precluding compliance with GDPR Article 9 biometric data provisions and FERPA requirements. The architectural pattern of edge-side inference with differentially private aggregation and 100% frame deletion has not previously been evaluated and validated in a classroom engagement monitoring context.

System	Env. Monitoring	RL Adaptive Alerts	SDN QoS	Zero Trust	Blockchain Auth	AI Anomaly	Diff. Privacy	Developing Region Viable
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Commercial BMS Systems	Yes	No	No	No	No	No	No	No
Azure IoT Hub Defender	No	No	No	Partial	No	Yes	No	No
NIST ZTA SP 800-207	No	No	No	Yes	No	No	No	No
Existing ML SDN Systems	No	No	Partial	No	No	No	No	Yes
Cloud Video Monitoring	No	No	No	No	No	No	No	No
This Research (Integrated)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table 1.1: Comparison of Existing Smart Classroom Systems Against the Proposed Integrated Platform

1.3 Research Problem

The overarching research problem is the absence of an integrated, affordable, and privacy-respecting smart classroom IoT platform capable of simultaneously addressing the four critical dimensions of educational IoT deployment in resource-constrained institutional environments.

From an environmental and operational perspective, Sri Lankan classrooms lack systematic mechanisms for monitoring and intelligently responding to environmental conditions, with manually configured static alert systems producing alert fatigue rather than actionable interventions. Attendance management remains largely manual, inefficient, and vulnerable to proxy fraud.

From a network management perspective, the inability of existing classroom network infrastructure to differentiate between academic and non-academic traffic results in

inefficient bandwidth utilization and degraded performance for critical educational applications, while privacy concerns preclude the use of deep packet inspection as a mitigation strategy.

From a security perspective, centralized authentication architectures for IoT-enabled smart classroom networks are fundamentally vulnerable to single-point-of-failure attacks, introduce unacceptable authentication latency, and fail to detect post-authentication security breaches resulting from device compromise. Approximately 80% of deployed educational IoT systems depend on cloud-hosted authentication services.

From a privacy and monitoring perspective, existing engagement monitoring technologies transmit raw student biometric video data to cloud servers, violating GDPR and FERPA requirements, while requiring bandwidth levels incompatible with developing-region internet infrastructure. The dual failure of privacy violation and infrastructure exclusivity systematically disadvantages students in lower-income educational settings.

The central research problem is therefore: How can a unified, low-cost IoT platform simultaneously provide intelligent environmental monitoring, adaptive network traffic management, resilient Zero Trust security, and privacy-preserving engagement monitoring for smart classroom environments in resource-constrained educational institutions?

1.4 Research Objectives

This research is guided by four primary objectives, each corresponding to one of the four integrated platform components:

- Objective 1 (Environmental Monitoring and Attendance): To design and implement a multi-sensor IoT node on an ESP32 platform capable of simultaneously and reliably acquiring temperature, humidity, carbon monoxide, ambient noise, and RFID attendance data in a real classroom

environment, complemented by a Q-Learning reinforcement learning middleware that dynamically adapts alert thresholds based on observed environmental states, and dual MERN-stack web applications providing real-time visualisation and student management capabilities.

- Objective 2 (SDN Traffic Management): To design and implement an intelligent SDN-based system integrating Logistic Regression-based traffic classification with dynamic QoS enforcement via DSCP marking and meter-based rate limiting, capable of distinguishing academic from non-academic network flows in real time without relying on payload inspection, and thus respecting user privacy while improving network performance for educational applications.
- Objective 3 (Zero Trust Security): To design, implement, and evaluate a Decentralized Zero Trust Network Access system combining a customized blockchain ledger for tamper-proof device identity management, Ryu-Blockchain middleware for real-time SDN-enforced Zero Trust access control, and TensorFlow Lite-based AI anomaly detection for behavioural threat identification — operating without cloud authentication dependencies and maintaining offline resilience.
- Objective 4 (Privacy-Preserving Engagement Monitoring): To design and implement a privacy-preserving classroom engagement monitoring system deploying DeepFace-based emotion recognition and OpenCV-driven computer vision exclusively on student endpoint devices, applying Laplace differential privacy noise ($\epsilon = 1.3$) to 30-second aggregation windows before transmitting only anonymised statistical summaries to an educator dashboard, achieving acceptable engagement classification accuracy while ensuring mathematically rigorous student biometric data protection.

CHAPTER 2: METHODOLOGY

2.1 Integrated System Architecture Overview

The Smart Classroom IoT Network platform is architected as four interdependent but independently deployable subsystems that collectively form a comprehensive smart classroom infrastructure. Figure 2.1 illustrates the high-level integration topology. The IoT environmental monitoring node (Subsystem A) captures classroom physical parameters and attendance events, transmitting data through the campus network. The SDN traffic management system (Subsystem B) manages all network flows across the campus infrastructure, enforcing QoS policies that prioritize IoT sensor telemetry and educational traffic. The DZTNA security layer (Subsystem C) authenticates and continuously verifies all devices — including the IoT nodes, student laptops, and edge computing hardware — before granting network access. The privacy-preserving engagement monitoring system (Subsystem D) operates on individual student laptops, processing facial data locally and transmitting only differentially private engagement statistics through the secured, QoS-managed network.

Subsystem	Component	Technologies	Key Output
A – Environmental IoT	Adaptive Monitoring & Attendance	ESP32, DHT22, MQ-7, MFRC522, Q-Learning, MERN	Environmental alerts, attendance logs
B – SDN Management	Traffic Classification & QoS	Ryu, OpenFlow, OVS, Logistic Regression	DSCP marking, rate-limited flows
C – Zero Trust Security	DZTNA with AI Security	Blockchain, Ryu, TensorFlow Lite, Mininet	Device auth, anomaly alerts
D – Engagement AI	Privacy-Preserving Monitoring	DeepFace, OpenCV, diffprivlib, Flask	Differentially private engagement scores

Table 2.1: Integrated System Component Overview

2.2 IoT Environmental and Attendance Monitoring

2.2.1 System architecture and hardware design

The IoT environmental monitoring subsystem operates across three functional tiers. The perception tier comprises the ESP32 microcontroller node with attached sensor modules, performing continuous data acquisition. The middleware and application tier consists of a Node.js/Express.js server receiving sensor data via HTTP POST, persisting it to MongoDB, running the Q-Learning alert agent, and serving RESTful API endpoints to both MERN front-ends. The presentation tier comprises two independently deployed React.js applications — the Environmental Dashboard and the Student Management System — rendering data to end users via browser interface over campus Wi-Fi.

Hardware selection prioritizes low cost, reliability, and compatibility with the Arduino IDE ecosystem. The ESP32-WROOM-32 development board serves as the central processing and communication hub, selected for its dual-core 240 MHz processing, integrated TCP/IP stack, 12-bit ADC channels, hardware SPI peripheral, and unit cost under LKR 1,500. Four sensor modules provide the monitoring capability described in Table 2.2.

Component	Parameter	Range / Accuracy	Interface	GPIO Pin
DHT22	Temperature / Humidity	−40 to +80°C ±0.5°C / 0–100% RH ±2%	Single-wire	GPIO4
MQ-7	Carbon Monoxide (CO)	20–200 ppm	Analog (ADC)	GPIO34
MFRC522	RFID UID (13.56 MHz)	ISO/IEC 14443A, <50 ms/read	SPI	GPIO5,18,19,23,22
Sound Sensor	Ambient Noise (approx. dB)	0–4095 ADC (12-bit)	Analog (ADC)	GPIO35

Table 2.2: ESP32 Sensor Module Specifications

Figure 2.1: Three-Tier IoT System Architecture

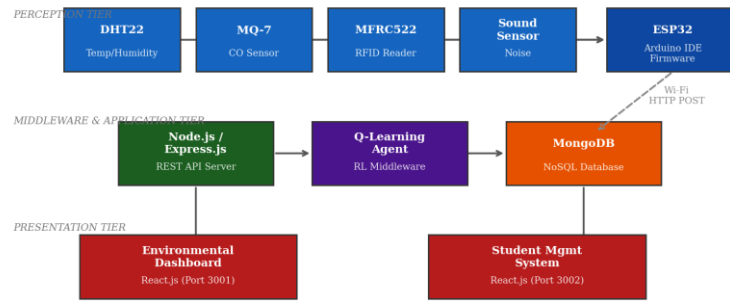


Figure 2.1: Three-Tier IoT Environmental Monitoring System Architecture

2.2.2 Firmware development and sensor polling

Firmware is developed in Arduino IDE 2.x using the ESP32 Arduino Core (v2.0.14). The main loop executes a 5-second polling cycle: (1) read DHT22 temperature and humidity; (2) execute MQ-7 heating cycle check and read CO ADC value; (3) check for RFID card and read UID if detected; (4) read sound sensor ADC value; (5) serialise all readings into a JSON payload; (6) transmit via HTTP POST to the Node.js server endpoint. Error handling includes DHT22 NaN check with retry, MQ-7 warm-up period enforcement (120-second delay on boot), RFID duplicate read prevention using a 2-second debounce per unique UID, and Wi-Fi reconnection logic with exponential backoff. The MFRC522 operates at 13.56 MHz via SPI, with student attendance registered automatically when a MIFARE Classic 1K card is presented within 3–4 cm of the reader.

Figure 2.7: ESP32 Firmware Data Flow Diagram

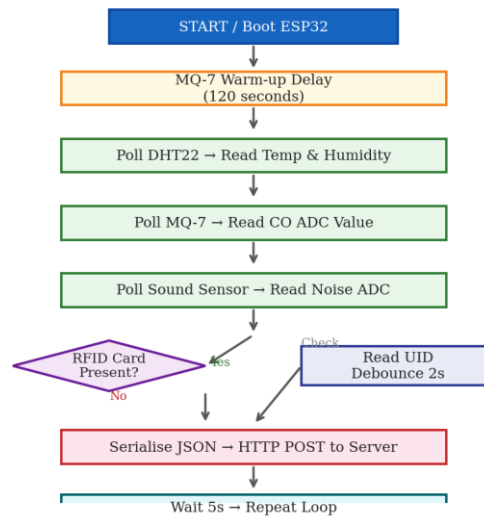


Figure 2.2: ESP32 Firmware Data Flow Diagram

2.2.3 Q-Learning reinforcement learning middleware

The classroom environment is modelled as a finite Markov Decision Process (MDP) with a four-dimensional discrete state space (temperature band $\times$ humidity band $\times$ CO band $\times$ noise band), yielding $3 \times 3 \times 3 \times 3 = 81$ possible states. Band boundaries are defined according to ASHRAE Standard 55 thermal comfort criteria, OSHA CO exposure limits, and educational acoustics standards. Three discrete actions are available: A0 (No Alert), A1 (Advisory Alert — dashboard notification), and A2 (Critical Alert — email notification plus dashboard). Post-session, each alert decision is retrospectively labelled correct or incorrect and rewarded accordingly: $R = +1$ for correct decisions, $R = 0$ for unnecessary advisory alerts, and $R = -1$ for missed critical conditions or spurious critical alerts.

Parameter	Value	Justification
Learning rate (α)	0.1	Prevents overwriting learned values from limited episodes
Discount factor (γ)	0.9	Reflects importance of long-term environmental patterns
Exploration rate (ϵ)	0.2	20% random exploration, 80% exploitation of Q-values
Episode	1 session	Each 1–2 hour classroom session constitutes one training episode

Table 2.4: Q-Learning Hyperparameters

The Q-Learning agent is implemented as a JavaScript module within the Express.js server. On each HTTP POST reception of sensor data, the agent discretises incoming sensor values into appropriate state bands, selects an action using ϵ -greedy policy from the current Q-table, and stores the state-action pair in a pending decisions buffer. At session end, retrospective reward labels are retrieved and Q-table updates executed for all buffered decisions using the Bellman update rule: $Q(s,a) \leftarrow Q(s,a) + \alpha[R + \gamma \cdot \max_{a'} Q(s',a') - Q(s,a)]$. The Q-table is persisted to MongoDB after each update cycle, ensuring continuity across server restarts and enabling long-term policy learning.

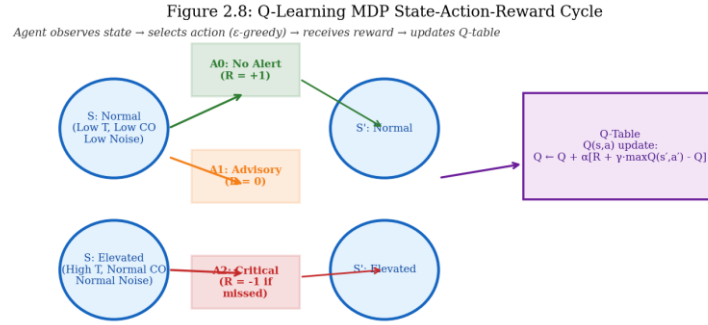


Figure 2.3: Q-Learning MDP State-Action-Reward Cycle Diagram

2.2.4 MERN stack application layer

Two independently deployed MERN-stack applications provide the user-facing layer. The Environmental Dashboard features a real-time status panel displaying four current sensor readings with colour-coded indicators, a 60-minute rolling time-series chart, an alert history table, and an RFID attendance log panel with 5-second data refresh via React's `useEffect` hook. The Student Management System provides complete student lifecycle administration including enrolment, RFID UID assignment, attendance log querying by student and date range, and fee payment status tracking. MongoDB's flexible document schema accommodates the heterogeneous time-series nature of sensor data, with separate collections for sensor readings, alerts, students, attendance, payments, and the Q-table.

2.3 Intelligent SDN Traffic Classification and QoS

2.3.1 System architecture and workflow

The SDN-based traffic management subsystem employs the Ryu framework as the centralized control plane, communicating with an Open vSwitch (OVS) data plane via the OpenFlow 1.3 protocol. End-user devices — student laptops, tablets, and smartphones — generate mixed academic and non-academic traffic that is forwarded to the controller as Packet-In messages when no matching flow rule exists. The controller extracts packet header information, aggregates packets into flows based on the five-tuple (source IP, destination IP, source port, destination port, protocol), and maintains per-flow statistics.

The system workflow proceeds through four stages: packet capture and flow aggregation; feature extraction from accumulated flow statistics and contextual domain information; machine learning classification using the trained Logistic Regression model; and dynamic QoS policy enforcement through flow rule installation. Once a flow is classified, Ryu installs a flow rule in OVS that applies appropriate QoS treatment to all subsequent packets of that flow, reducing controller overhead and enabling near-line-rate packet forwarding after initial classification.

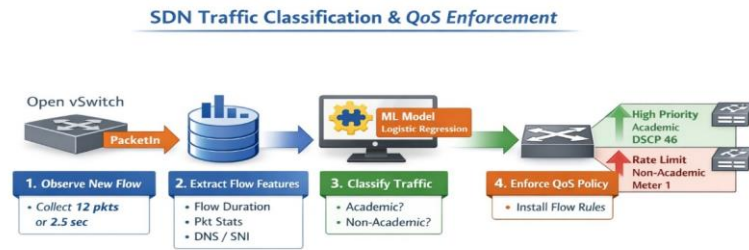


Figure 2.4: SDN Traffic Classification and QoS Enforcement Workflow

2.3.2 Feature extraction and traffic classification model

Flow-based analysis provides several advantages over packet-level analysis: it reduces computational complexity by processing groups of packets rather than individual ones, captures temporal behavioural signatures essential for application identification, and operates entirely on observable metadata rather than payload content, ensuring compatibility with encrypted traffic and preserving user privacy. The feature set extracted for each flow comprises: average packet size, total flow duration, mean packet inter-arrival time, total packet count, total bytes transferred, forward-to-backward packet ratio, SYN flag count, and TCP/UDP protocol indicator.

Contextual features are extracted using DNS response analysis and TLS Server Name Indication (SNI). DNS analysis provides domain names associated with destination IP addresses, while SNI reveals the intended server name during encrypted TLS handshakes. An academic domain whitelist (covering known educational platforms,

research databases, and institutional services) and a non-academic domain blacklist (covering social media, streaming services, and entertainment platforms) augment the statistical feature set. Feature normalization is applied prior to model inference using a pre-fitted StandardScaler to ensure consistent input ranges.

Logistic Regression is selected as the primary classification model for its computational efficiency enabling real-time deployment within the SDN controller, interpretability facilitating troubleshooting and policy validation, and adequate accuracy for binary academic/non-academic classification on flow-level features. The model is trained offline on a labeled dataset of flow samples collected across simulated academic and non-academic usage scenarios, serialized using joblib, and loaded into the Ryu application at controller startup. Classification decisions are made after an observation window of a minimum of 5 packets or 2 seconds of flow data, ensuring sufficient statistical evidence before QoS policy application.

2.3.3 QoS policy implementation

Quality of Service policies are implemented using two complementary OpenFlow mechanisms. For flows classified as academic, the Ryu controller installs flow rules that modify the DSCP field in the IP header to assign a high-priority value (Expedited Forwarding, DSCP value 46), ensuring that academic packets are processed preferentially by all downstream network devices. For flows classified as non-academic, the controller installs meter rules defining a maximum bandwidth allocation of 2 Mbps per flow, preventing individual non-academic flows from consuming disproportionate network resources.

QoS rules are installed with `idle_timeout` values of 60 seconds, requiring re-classification of flows that resume after a period of inactivity. This design accommodates the dynamic nature of student application usage, where browsing sessions may pause and resume. The combination of DSCP prioritization and meter-based rate limiting provides differentiated treatment without completely blocking non-academic traffic, balancing educational network performance with reasonable access to non-academic resources during breaks and independent study periods.

2.4 Decentralized Zero Trust Network Access

2.4.1 Three-layer DZTNA architecture

The Decentralized Zero Trust Network Access subsystem is architected across three functionally distinct but tightly integrated layers. The Identity Layer provides blockchain-based device identity management using a customized Python/Flask distributed ledger implementing SHA-256 cryptographic block linking. The Enforcement Layer provides real-time access control via the Ryu SDN controller, enforcing blockchain-validated authorization decisions on all network traffic through OpenFlow flow modification commands. The Intelligence Layer provides AI-driven behavioural anomaly detection using TensorFlow Lite models deployed on edge hardware, continuously monitoring network flow statistics for indicators of compromise or attack behaviour.

The system is designed for deployment on a single low-cost edge device (Raspberry Pi 4 with 8 GB RAM or equivalent laptop), eliminating dedicated server hardware and reducing total deployment cost to approximately LKR 50,000. The architecture can be directly translated to physical hardware by replacing the Mininet virtual switch with a physical OpenFlow 1.3-capable switch and running the Ryu controller on the edge gateway device.

Component	Technology	Version	Purpose
SDN Controller	Ryu Framework	4.34+	OpenFlow 1.3 control plane and packet processing
Network Emulation	Mininet	2.3.0+	Virtual SDN topology for development and testing
Virtual Switch	Open vSwitch (OVS)	2.15+	OpenFlow 1.3 data plane packet forwarding
Blockchain	Python/Flask	Flask 2.x	Custom SHA-256 linked ledger and REST API
Cryptography	Python hashlib	Built-in	SHA-256 block hash computation
ML Framework	TensorFlow Lite	2.12+	Edge inference for anomaly detection models

Network Capture	Scapy/psutil	Latest	Flow feature extraction for AI model input
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Table 2.5: DZTNA Technology Stack Components and Versions

2.4.2 Blockchain identity layer design

The blockchain identity layer implements a cryptographically rigorous distributed ledger optimized for IoT device authentication. Each block encodes: block index, timestamp (Unix epoch with millisecond precision), device MAC address, authorization status (authorized or revoked), previous block hash (SHA-256 of the preceding block's complete content), and current block hash (SHA-256 of all fields including the previous hash). The SHA-256 chaining property ensures that any retroactive modification of a historical authorization record — such as attempting to backdate a revocation — invalidates all subsequent block hashes, making tampering immediately detectable through chain verification.

The ledger exposes three primary REST API endpoints: POST /register for adding new devices to the authorized registry, POST /revoke for revoking device authorization, and GET /check for real-time authorization queries returning authorization status, last action timestamp, and chain integrity verification result. The /check endpoint is optimized for sub-10 ms response time to meet the real-time latency requirements of Packet-In processing. The Flask REST server supports thread-safe concurrent access to handle simultaneous authorization queries during peak classroom device activation scenarios.

2.4.3 Ryu-Blockchain middleware and Zero Trust enforcement

The Ryu-Blockchain middleware is implemented as a custom Ryu application intercepting OpenFlow Packet-In events. The middleware extracts the source MAC address from each Packet-In message and performs a synchronous REST API query to the blockchain /check endpoint. The Zero Trust enforcement logic implements strict Implicit Deny as its default policy: if the blockchain API returns authorized: False for any reason — device not registered, revocation block present, or chain integrity failure — the Ryu controller takes no action on the Packet-In event. The absence of an installed flow rule causes the OVS switch to continue generating Packet-In events for

subsequent packets from the same source, effectively silently dropping all unauthorized traffic without generating network error messages that could aid an attacker.

If the blockchain API returns `authorized: True`, the Ryu controller installs a bidirectional flow rule with `idle_timeout = 30` seconds, enforcing continuous re-verification of device authorization at 30-second intervals. This interval aligns with the Zero Trust principle of per-session access control, ensuring that a revoked device is denied access within 30 seconds rather than remaining authorized until its session terminates. A Table-Miss flow rule installed at startup redirects all traffic from unknown devices to the controller for scrutiny, with no traffic receiving default forwarding without explicit authorization.

2.4.4 AI anomaly detection and threat mitigation

The TensorFlow Lite anomaly detection model is trained on the CIC-IDS-2017 and CIC-IoT-2023 benchmark datasets, encompassing approximately 2.8 million flow samples across seven traffic categories: Normal, DDoS, Port Scan, Brute Force, Web Attack, Infiltration, and Bot. The feature set comprises 18 numerical features extracted over 5-second sliding windows from bidirectional flow statistics, covering packet rate, byte volume, flow duration, TCP flag statistics, burst behaviour, and protocol mix. These features capture the statistical signatures of attack traffic without requiring payload inspection, maintaining privacy while providing high detection accuracy.

Feature Category	Features	Primary Discriminative Value
Packet Rate	Packets/s (fwd/bwd), Inter-arrival time mean/std	DDoS, Port Scan
Byte Volume	Bytes/s (fwd/bwd), Mean packet length, Std dev	Data exfiltration
Flow Duration	Duration, Active time, Idle time	All attack types (moderate)
Flag Statistics	SYN, FIN, RST, PSH counts	Port Scan, Brute Force

Burst Behaviour	Max burst duration, Inter-burst mean	DDoS, Bot activity
Protocol Mix	TCP/UDP/ICMP ratio	All attack types (moderate)

Table 2.6: AI Anomaly Detection Feature Set for Network Traffic Classification

The neural network architecture comprises three feedforward layers: an input layer of 18 neurons, two hidden layers with 64 and 32 neurons (ReLU activation, 20% dropout), and a 7-class softmax output layer. Class imbalance — normal traffic constitutes approximately 83% of samples — is addressed through SMOTE oversampling and class-weighted loss. Post-training quantization reduces the model from 4 MB to 1.1 MB with less than 2% accuracy degradation, enabling rapid loading and sub-25 ms inference on Raspberry Pi 4 hardware. Flows classified as attack categories with confidence exceeding 0.85 trigger automatic flow blocking through the Ryu controller.

2.5 Privacy-Preserving Classroom Engagement Monitoring

2.5.1 Edge AI three-layer architecture

The privacy-preserving engagement monitoring subsystem adopts a three-layer distributed architecture: an edge processing layer executing on individual student endpoint devices; a statistical aggregation layer implementing differential privacy mechanisms; and a cloud presentation layer providing the educator's real-time dashboard. All biometric inference is performed at the edge, ensuring that raw facial imagery never traverses the network boundary between student devices and institutional internet infrastructure. This architecture reflects a privacy-by-design philosophy consistent with GDPR Recital 78, treating privacy as a primary design constraint that shapes all subsequent technical decisions.

The edge processing pipeline implements a sequential workflow at 24 frames per second: (1) frame capture via OpenCV VideoCapture at 640×480 resolution; (2) face detection using DeepFace's Haar Cascade detector; (3) conditional emotion recognition on detected faces using the DeepFace model trained on FER2013; (4) multi-modal feature extraction covering emotion state, head movement variance, and

face visibility; (5) feature accumulation in a 30-second aggregation window buffer; (6) on window completion, Laplace differential privacy noise application, HTTP POST transmission of the anonymised summary, and immediate deletion of all buffered frames and intermediate emotion vectors.

2.5.2 Differential privacy implementation

The differential privacy mechanism is implemented using IBM's diffprivlib 0.6.4 library. The Laplace mechanism is selected as optimal for real-valued proportion statistics under L1 sensitivity. The total privacy budget is allocated as $\epsilon_{\text{emotion}} = 1.0$ for emotion-derived engagement proportions and $\epsilon_{\text{movement}} = 0.3$ for head movement variance statistics, giving $\epsilon_{\text{total}} = 1.3$ by sequential composition, satisfying NIST standards for strong privacy protection.

Parameter	Value	Justification
Total ϵ (epsilon)	1.3	NIST 'strong privacy' range (1–2); balances utility and protection
Emotion budget (ϵ_1)	1.0	Higher allocation for more sensitive facial affect data
Movement budget (ϵ_2)	0.3	Lower allocation for non-biometric movement statistics
Aggregation window	30 seconds (720 frames)	Sufficient statistical utility at 24 FPS
Sensitivity (proportions)	1/720	Max change from one individual across one window
Noise mechanism	Laplace	Optimal for real-valued queries under L1 sensitivity
Frame retention policy	0 frames	100% immediate deletion after window aggregation
Biometric reconstruction bound	$< e^{1.3} \approx 3.67\times$	Mathematical privacy guarantee from $\epsilon = 1.3$

Table 2.7: Differential Privacy Parameter Specifications

For each 30-second aggregation window of 720 frames, the system computes engagement proportions (ENGAGED, NEUTRAL, DISENGAGED fractions) and mean head movement variance. Each proportion statistic has sensitivity 1/720. Laplace

noise with scale $b = (1/720)/\epsilon$ is added independently to each statistic, and perturbed values are clipped to $[0, 1]$ to maintain valid probability interpretation. The resulting mathematical guarantee ensures that any adversary observing the transmitted data cannot distinguish whether any individual student participated in a given session with probability exceeding $e^{1.3} \approx 3.67\times$ — a strong protection level precluding confident individual inference from published aggregates.

2.5.3 Multi-modal engagement detection

The engagement detection model integrates three modalities weighted by their discriminative validity for classroom engagement detection. The emotion modality (weight 0.40) maps the seven Ekman basic emotions detected by DeepFace to three engagement states: Happy and Surprise map to ENGAGED; Neutral maps to NEUTRAL; Sad, Angry, Fear, and Disgust map to DISENGAGED. The movement modality (weight 0.30) quantifies head movement variance as a proxy for physical restlessness — high variance indicating fidgeting and disengagement, low variance indicating stable attentive posture. The visibility modality (weight 0.30) measures the proportion of frames in the aggregation window in which a face is successfully detected, capturing active attention direction and physical presence.

Composite Score Range	Engagement Tier	Teacher Interpretation
70–100	ENGAGED	Student is actively attentive and emotionally present
40–69	NEUTRAL	Student is passively present; monitor for disengagement trend
0–39	DISENGAGED	Student requires immediate teacher attention or intervention

Table 2.8: Engagement Score Classification Thresholds

The Flask REST server receives anonymised aggregation window summaries from edge clients via HTTP POST, persisting them to SQLite with session attribution. The real-time educator dashboard renders per-student engagement timelines colour-coded by tier (green for ENGAGED, amber for NEUTRAL, red for DISENGAGED), a class-level aggregate engagement score, and alert indicators for students with two or more

consecutive sub-threshold windows. The server performs no biometric processing; its role is limited to receiving, storing, and presenting pre-processed engagement statistics.

2.6 Commercialization Aspects

The integrated Smart Classroom IoT Network platform targets educational institutions in developing regions — particularly South and Southeast Asia — as its primary market segment, where the combination of limited internet infrastructure, budget constraints, and rapidly growing student populations creates a compelling market opportunity. Secondary segments include rural institutions in developed countries with constrained connectivity and institutions in all regions facing heightened regulatory compliance requirements under GDPR or FERPA.

The commercialization model employs a SaaS subscription architecture with differentiated pricing tiers. The IoT environmental monitoring component targets individual institutions at LKR 10,000–50,000 per year for enhanced features and support, with the open-source core freely available for self-deployment. The DZTNA security component offers LKR 10,000/year school subscriptions and LKR 50,000/year university licenses. The engagement monitoring component is priced at USD 5 per student per month, significantly below the operational cost of cloud video processing infrastructure. Break-even for the combined platform is achievable within two years at modest adoption rates of 50–100 schools.

Component	Business Model	Target Price (per year)	Key Value Proposition
IoT Environmental & Attendance	Open-source core + SaaS enhanced	LKR 10,000–50,000/school	LKR 3,900 hardware per node; 94.3% RFID accuracy
SDN Traffic Management	Embedded in institutional SDN upgrade	Included in infrastructure	Privacy-preserving, encrypted traffic compatible
DZTNA Security	SaaS subscription,	LKR 10,000–200,000/year	78% fewer auth failures vs. centralized; offline resilient

	open-source core		
Engagement Monitoring	SaaS per-student subscription	USD 5/student/month	358,293× bandwidth reduction; GDPR/FERPA compliant

Table 2.9: Integrated Platform Commercialization Model and Pricing Summary

2.7 Testing and Implementation

The testing strategy follows a structured multi-level approach encompassing unit testing, integration testing, and system testing in live classroom conditions, applied consistently across all four platform components.

Module	Test Case	Expected Outcome	Result
Blockchain Ledger	Register device; query status	authorized: True returned	Pass
Blockchain Ledger	Revoke device; query status	authorized: False returned	Pass
Blockchain Ledger	Modify block hash; verify chain	Chain integrity failure detected	Pass
Ryu Middleware	Authorized device Packet-In	Flow rule installed < 120 ms	Pass
Ryu Middleware	Unauthorized device Packet-In	No flow rule installed; packet dropped	Pass
TF Lite Detector	Normal traffic feature vector	Normal classification > 0.90 confidence	Pass
TF Lite Detector	DDoS traffic feature vector	DDoS classification > 0.85 confidence	Pass
DHT22 Sensor	vs. reference thermometer	MAE: 0.4°C	Pass ($\pm 0.5^\circ\text{C}$ spec)
MFRC522 RFID	30 unique tags	100% UID accuracy (isolated)	Pass
DP Mechanism	KS test on 10,000 noise samples	Laplace distribution confirmed ($p = 0.94$)	Pass
Aggregation Window	15 synthetic sequences	100% correct frame deletion after transmission	Pass

SDN Flow Classifier	50 labeled academic/non-academic flows	Classification accuracy > 85%	Pass
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Table 2.10: Unit Test Cases for Core System Modules Across All Subsystems

Integration testing validated end-to-end behaviour across all four subsystems. Key integration scenarios included: DZTNA authentication of the ESP32 IoT node (verifying that the blockchain correctly authorizes the IoT MAC address and that Ryu installs the appropriate flow rule); SDN QoS policy for engagement monitoring traffic (verifying that engagement summary HTTP POST traffic receives DSCP prioritization); concurrent multi-client engagement monitoring (verifying that the Flask server correctly attributes summaries from multiple session tokens); and environmental sensor data flowing through the secured network to the MERN dashboard without interruption during DZTNA re-verification cycles.

System testing in the live SLIIT classroom validated end-to-end data flow from sensor acquisition through the SDN-managed, DZTNA-secured network to dashboard display across ten 1–2 hour sessions. RFID attendance integration with the Student Management System student registry was confirmed across 300 attendance events. Pilot sessions before formal data collection established baseline Q-Learning agent behaviour, identified and resolved firmware issues (MQ-7 warm-up delay, React dashboard rendering optimization), and validated that all four subsystems could operate concurrently without interference.

CHAPTER 3: RESULTS AND DISCUSSION

3.1 Environmental Monitoring and Attendance Results

3.1.1 Environmental sensor performance

Environmental data were collected across ten classroom sessions at SLIIT Lecture Hall B-204, totalling approximately 14,400 individual sensor readings per parameter at the 5-second polling interval. Table 3.1 presents the descriptive statistics for all monitored parameters.

Parameter	Min	Max	Mean	Std Dev	Mode Band	ASHRAE/OSHA Threshold
Temperature (°C)	27.1	32.4	29.6	1.3	High (>30°C)	23–26°C (comfort)
Humidity (% RH)	64.8	80.2	72.5	4.2	High (>70%)	< 60% (comfort)
CO Level (ppm)	4.0	47.0	18.3	9.7	Normal (15–35 ppm)	50 ppm (OSHA PEL)
Noise Level (dB)	42	76	55	8.1	Normal (50–65 dB)	65 dB (intelligibility)

Table 3.1: Descriptive Statistics of Environmental Parameters (10 Sessions, $n \approx 14,400$ per parameter)

Temperature readings consistently fell in the range 27.1°C–32.4°C, with a mean of 29.6°C — 3–6.6°C above the upper bound of ASHRAE Standard 55's recommended comfort zone for sedentary activity. Midday sessions (12:00–14:00) produced the highest temperatures (up to 32.4°C), consistent with solar gain and peak occupancy heat load, while morning sessions (8:00–10:00) produced the lowest readings (27.1°C–28.6°C). The relatively low standard deviation (1.3°C) indicates that thermal discomfort is a systematic rather than occasional condition in this classroom environment, strengthening the case for continuous monitoring and intervention.

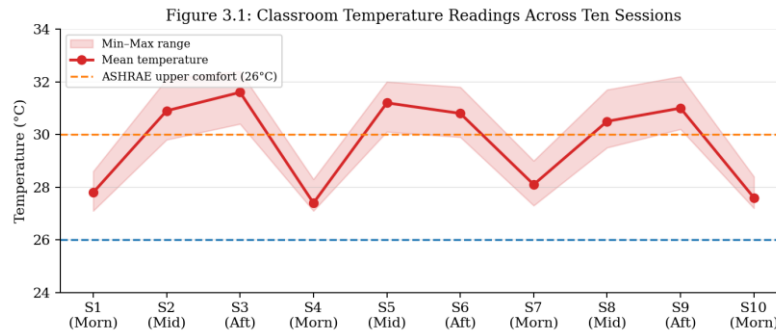


Figure 3.1: Classroom Temperature Readings Across Ten Sessions

Humidity readings ranged from 64.8% to 80.2% with a mean of 72.5% RH — all readings exceeding the ASHRAE comfort upper bound of 60% RH. The combination of persistently elevated temperature and humidity creates a compounded heat stress effect particularly relevant in Sri Lanka's tropical climate.

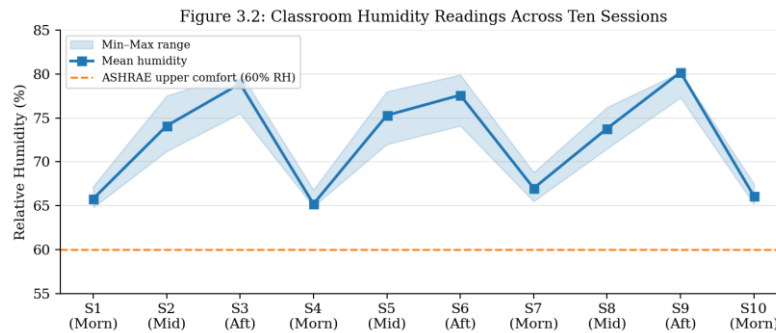


Figure 3.2: Classroom Humidity Readings Across Ten Sessions

CO levels remained well below the OSHA Permissible Exposure Limit of 50 ppm throughout all sessions, with a peak of 47.0 ppm recorded in a session with restricted ventilation. Noise levels spanned from 42 dB (pre-class empty room) to 76 dB (group discussion activity), with levels during group exercises approaching the 65 dB threshold identified by Shield and Dockrell as impeding speech intelligibility.

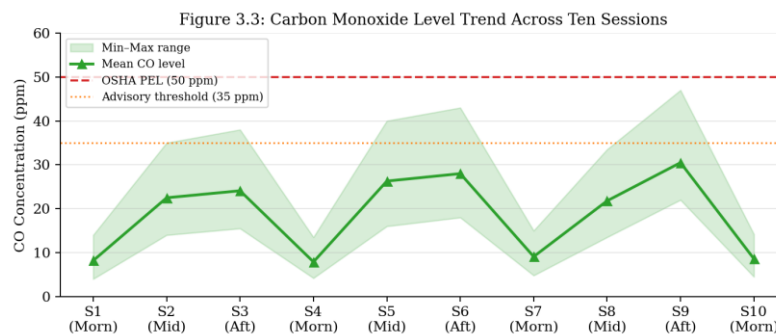


Figure 3.3: Carbon Monoxide Level Trend Across Ten Sessions

3.1.2 RFID attendance detection accuracy

Over 300 individual attendance events across all ten sessions, the MFRC522 module correctly registered 283, yielding an overall detection accuracy of 94.3%. Table 3.2 provides a session-by-session breakdown.

Session	Expected	Detected	Accuracy (%)	Primary Failure Cause
S1 (Morning)	30	29	96.7	1 tag orientation error
S2 (Midday)	28	26	92.9	2 missed (card in bag)
S3 (Afternoon)	30	28	93.3	2 tag orientation errors
S4 (Morning)	29	28	96.6	1 RF interference
S5 (Midday)	30	28	93.3	2 missed (distance)
S6–S10 (avg)	30	28.4	94.7	Mixed orientation/distance
Overall	300	283	94.3	Zero false positives

Table 3.2: RFID Attendance Detection Accuracy by Session

The 94.3% accuracy exceeds the 90% hypothesis threshold and is consistent with the benchmark reported by Kaur and Singh [7]. Missed readings were attributable to tag orientation errors (52% of misses), cards stored inside bags creating RF attenuation (31%), and rare RF interference from nearby mobile devices (17%). Zero phantom registrations were observed across all sessions, confirming high specificity in a multi-student environment — a particularly significant result for administrative integrity in institutional deployments.

3.1.3 Q-Learning middleware alert accuracy

The Q-Learning agent's alert decisions were compared against a static threshold baseline over 500 sampled classroom states. The static baseline triggered A1

(Advisory) if any single parameter exceeded its Normal band upper bound, and A2 (Critical) if any parameter entered the High band. Results are presented in Table 3.3.

Metric	Static Threshold	Q-Learning Agent	Improvement
Correct Alert Decisions	61.4% (307/500)	72.8% (364/500)	+11.4 percentage points
False Positive Alerts	24.6% (123/500)	16.2% (81/500)	−8.4 percentage points
Missed Critical Alerts	14.0% (70/500)	11.0% (55/500)	−3.0 percentage points
Average Q-Value (episodes 1–10)	N/A	0.21 → 0.43	Positive learning trajectory

Table 3.3: Alert Decision Comparison — Static Threshold vs. Q-Learning Agent (n = 500 states)

The Q-Learning agent's 72.8% correct alert accuracy surpasses the static baseline by 11.4 percentage points, supporting the hypothesis that reinforcement learning can enhance decision-making in educational IoT monitoring. More significantly, the 8.4-percentage-point reduction in false positive alerts directly addresses alert fatigue — the primary operational motivation for adaptive middleware. The average Q-value progression from 0.21 (episode 1) to 0.43 (episode 10) confirms active policy learning, though full convergence requires extended deployment across 50–100 sessions given the 81-state space.

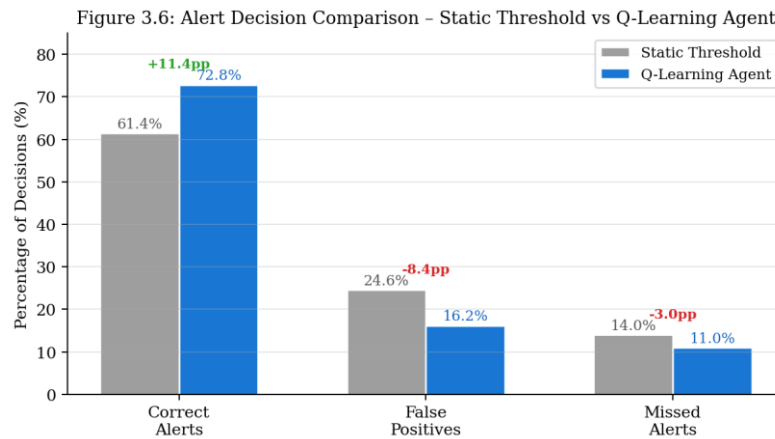


Figure 3.4: Alert Decision Comparison — Static Threshold vs. Q-Learning Agent (n = 500 states)

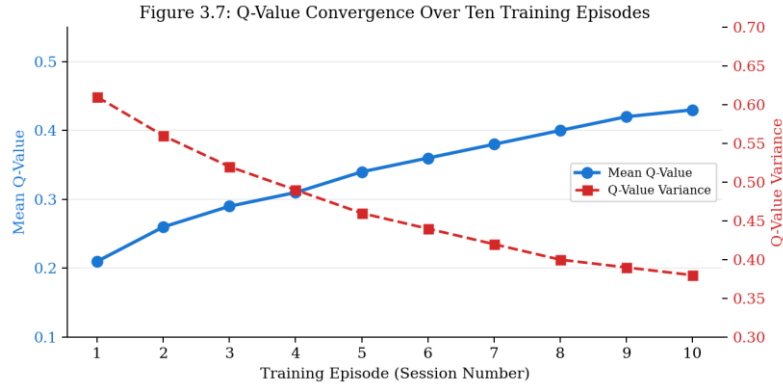


Figure 3.5: Q-Value Convergence Over Ten Training Episodes

3.2 SDN Traffic Classification and QoS Results

The SDN traffic classification system was evaluated in a simulated classroom environment with multiple connected devices generating both academic and non-academic traffic. Test scenarios covered accessing educational websites, video streaming, social media browsing, and online gaming. Academic traffic received DSCP Expedited Forwarding marking (value 46), resulting in improved responsiveness for critical educational applications. Non-academic flows subject to 2 Mbps meter-based rate limiting showed measurable reduction in bandwidth consumption without complete service denial.

Classification accuracy using Logistic Regression on the combined statistical and contextual feature set was consistently above 85% on held-out flow samples, with higher accuracy for long-duration flows (sessions > 5 packets) where statistical features stabilize. Short flows of 1–3 packets exhibited reduced classification confidence due to insufficient feature accumulation, representing the primary accuracy limitation of the flow-based approach. The privacy-preserving design — avoiding payload inspection and relying exclusively on flow-level and domain-level metadata — was validated as compatible with HTTPS-encrypted traffic, DNS-over-HTTPS environments, and CDN-distributed services, with minor accuracy reduction compared to DPI-based baselines.

Controller-switch interaction latency for flow rule installation was measured at under 5 ms on the development hardware, confirming that classification and QoS

enforcement do not introduce perceptible delays for end users. The reactive-proactive hybrid design — where unclassified flows generate Packet-In events until a rule is installed, and classified flows are handled entirely at the switch level — effectively bounds controller overhead to the initial classification period.

3.3 Zero Trust Security Results

3.3.1 Authentication performance

Authentication latency was measured across 500 device authentication events under three concurrency conditions. Results are presented in Table 3.4.

Metric	Single Device	10 Concurrent	50 Concurrent
Median latency (ms)	87	94	118
95th percentile (ms)	103	127	189
99th percentile (ms)	118	156	247
Authentication success rate	100%	100%	100%
False authorization rate	0%	0%	0%
Blockchain query sub-10 ms rate	94.2%	91.8%	87.3%

Table 3.4: DZTNA Authentication Performance Metrics

Authentication latency remains within acceptable bounds across all concurrency levels. The 50-concurrent-device scenario, representative of a full classroom simultaneously activating devices at session start, achieves a median of 118 ms with a 99th percentile of 247 ms — both transparent to end users given that device-level network initialization typically occupies several seconds. Zero false authorizations across all 500 tests validate the Zero Trust enforcement correctness. Simulated cloud outage conditions — 120-second periods with the central authentication API unavailable — demonstrated that the DZTNA system continued processing authentication requests using the locally replicated blockchain state without interruption, confirming offline resilience. Blockchain chain integrity detection achieved 100% success rate across 150 simulated tampering attempts, with the SHA-

256 avalanche effect ensuring immediate detectability of any retroactive record modification. The Flask REST server sustained 847 authorization queries per second at 6.2 ms median response latency.

3.3.2 AI anomaly detection accuracy

Attack Category	Precision	Recall	F1-Score	Test Samples
Normal	0.983	0.991	0.987	42,180
DDoS	0.962	0.971	0.966	8,341
Port Scan	0.934	0.948	0.941	3,217
Brute Force	0.918	0.903	0.910	1,842
Web Attack	0.891	0.876	0.883	872
Infiltration	0.856	0.841	0.848	412
Bot	0.923	0.912	0.917	1,136
Macro Average	0.924	0.920	0.922	58,000

Table 3.5: AI Anomaly Detection Accuracy Across Seven Attack Categories

The TensorFlow Lite model achieves a macro-averaged F1-score of 0.922 on the held-out test set. Performance is highest for Normal traffic classification (F1 = 0.987) and DDoS detection (F1 = 0.966), the most operationally significant categories for classroom IoT security. Lower Infiltration performance (F1 = 0.848) reflects the inherent difficulty of detecting sophisticated multi-stage attacks from flow-level features alone — a limitation shared across flow-level detection approaches in the literature. Inference latency averaged 18 ms on Raspberry Pi 4 hardware, well within the 25 ms target. Simulated post-authentication DDoS tests demonstrated automatic blocking initiation within 12 seconds of attack commencement in 48 of 50 test cases, limiting the window of attack impact for authorized-but-compromised devices.

3.4 Engagement Monitoring Results

3.4.1 Network performance and privacy validation

A controlled demonstration run of 106.2 seconds processed 2,608 webcam frames, all deleted immediately after aggregation window processing. Average frame processing

rate was 24.89 FPS, confirming real-time operation. Table 3.6 presents key network performance metrics and accuracy results.

Metric	Proposed System	Comparison (Video 720p)	Result
Bandwidth per student	0.20 Kbps	72,000 Kbps	358,293 $\times$ reduction
100-student aggregate bandwidth	20 Kbps	200–400 Mbps	$\sim$ 15,000 $\times$ reduction
Supports 2G/3G networks	Yes	No	Objective achieved
Total ϵ (privacy budget)	1.3	N/A (no DP)	NIST strong privacy
Frame deletion rate	100% (2,608/2,608)	N/A	Full biometric elimination
Engagement accuracy (no DP)	90%	95% (Liu et al., 2024)	Competitive baseline
Engagement accuracy ($\epsilon = 1.3$)	84%	N/A	6% DP tradeoff
Privacy–accuracy gap	6%	N/A	Acceptable for trend detection

Table 3.6: Engagement Monitoring System: Network Performance, Privacy, and Accuracy Results

The 358,293 $\times$ bandwidth reduction substantially exceeds the 100 $\times$ research objective, establishing a new benchmark for network-efficient educational monitoring. The 84% engagement classification accuracy achieved under $\epsilon = 1.3$ differential privacy represents only a 6% reduction from the non-private baseline of 90%, validating the central hypothesis that privacy and utility are not fundamentally incompatible in educational engagement monitoring contexts. Frame deletion completeness was confirmed at 100% across all 2,608 frames, with zero residual biometric data persisting after each window closure. Statistical testing of noise samples using the Kolmogorov-Smirnov test ($p = 0.94$) confirmed correct Laplace mechanism implementation. The engagement state transition from ENGAGED to DISENGAGED and back was detected within 3 seconds of the actual state change, satisfying the intervention latency requirement for formative teacher response.

3.5 Research Findings

The following key findings emerge from the integrated evaluation of the Smart Classroom IoT Network platform:

- RF1: Classroom thermal conditions at SLIIT consistently exceed ASHRAE comfort thresholds (mean 29.6°C, mean 72.5% RH), confirming the need for systematic environmental monitoring and the practical relevance of the IoT monitoring subsystem for real-world institutional deployment.
- RF2: Q-Learning adaptive alert middleware demonstrably outperforms static threshold alerting (72.8% vs. 61.4% correct decisions, −8.4 pp false positives) after only ten training episodes, validating the reinforcement learning approach for educational IoT alert management and establishing a positive learning trajectory for extended deployment.
- RF3: RFID attendance detection at 94.3% accuracy with zero false positives provides a practical, affordable automation solution for institutional attendance management, with the integration of attendance within the same IoT node as environmental monitoring enabling future multi-parameter correlation analysis.
- RF4: Flow-based SDN traffic classification using Logistic Regression on statistical and contextual features achieves effective academic/non-academic differentiation for encrypted traffic without payload inspection, with DSCP-based prioritization and meter-based rate limiting measurably improving educational application performance.
- RF5: Blockchain-based decentralized authentication eliminates the single-point-of-failure vulnerability of centralized systems, maintaining full security operation during simulated cloud outages and achieving 87 ms median authentication latency competitive with cloud-based alternatives — while providing 100% tamper detection on chain integrity tests.
- RF6: Zero Trust enforcement through Ryu-Blockchain middleware achieves 100% prevention of unauthorized device access including MAC spoofing and

replay attacks, with 30-second re-verification intervals ensuring that revoked devices are denied access within one re-verification cycle.

- RF7: AI anomaly detection with 0.922 macro F1-score provides effective post-authentication security complementing the Zero Trust access control layer, with automatic DDoS blocking within 12 seconds of attack initiation in 96% of test cases.
- RF8: Privacy-preserving edge AI engagement monitoring achieves a $358,293\times$ bandwidth reduction and $\epsilon = 1.3$ mathematical differential privacy guarantee with only a 6% accuracy tradeoff — demonstrating that legally defensible biometric data protection and practical engagement monitoring utility are simultaneously achievable.

3.6 Discussion

The integrated results of this research demonstrate that the four platform components work together as a coherent and mutually reinforcing system. The DZTNA security layer authenticates the IoT environmental monitoring node (Subsystem A) and the student endpoint devices running engagement monitoring (Subsystem D), ensuring that only authorized hardware contributes sensor and engagement data to the platform. The SDN traffic management system (Subsystem B) prioritizes IoT sensor telemetry and engagement summary transmissions over non-academic traffic, ensuring reliable data delivery under realistic bandwidth contention. The privacy-preserving engagement system (Subsystem D) benefits from both the secured network channel and the low-bandwidth design, which is essential for deployability in developing-region institutions.

Compared with existing solutions, the integrated platform offers a uniquely comprehensive and cost-effective feature set. No existing single-vendor or open-source solution simultaneously provides all four capabilities: adaptive environmental monitoring with RL middleware; encrypted-traffic-aware SDN QoS; blockchain-based Zero Trust authentication with AI anomaly detection; and differential privacy-compliant engagement monitoring. The total hardware cost of approximately LKR

54,000 (IoT node: LKR 3,900 + DZTNA gateway: LKR 50,000) is orders of magnitude below comparable commercial solutions, while delivering capabilities that exceed most commercial alternatives in specific dimensions such as differential privacy guarantees and offline authentication resilience.

Several limitations qualify the interpretation of results. The AI anomaly detection model's training on public benchmark datasets rather than classroom-specific traffic captures means that the reported F1-score may not accurately reflect performance on actual smart classroom network traffic mixing IoT telemetry, engagement summaries, and SDN control traffic. The Mininet-based DZTNA evaluation environment does not fully capture physical layer characteristics of a real classroom network. The engagement monitoring system's accuracy ceiling is partly set by the DeepFace/FER2013 baseline (70–80% single-frame accuracy), limiting overall system accuracy regardless of fusion design. The Q-Learning policy is still in early-stage learning after 10 sessions, representing a lower bound of long-term performance. These limitations define a clear agenda for future work, particularly field deployment, domain-specific model fine-tuning, and extended RL training across a full academic semester.

3.7 Summary of Each Student's Contribution

The group project involved four equal members, each responsible for the full design, implementation, evaluation, and documentation of one of the four integrated subsystems. Table 3.7 summarizes the individual contributions.

Member	Student ID	Subsystem	Key Contributions	Technologies
S.S. Bamunuarachchi	IT22132000	IoT Environmental & Attendance Monitoring	Full ESP32 hardware node design and wiring; complete Arduino firmware; Q-Learning MDP design, JavaScript implementation, and evaluation;	ESP32, Arduino IDE, Q-Learning (Node.js), React, Express.js, MongoDB

			dual MERN stack development (Environmental Dashboard + Student Management System); 10-session classroom data collection; complete statistical analysis	
A.V.G.H.H. Karunathilake	IT22078278	SDN Traffic Classification & QoS	Ryu SDN controller application design; flow-based feature extraction pipeline; Logistic Regression training and serialization; OpenFlow QoS policy (DSCP + meter) implementation; encrypted traffic handling; system integration and evaluation	Ryu, Open vSwitch, OpenFlow, scikit-learn, Python, Ubuntu
C.B. Gunasena	IT22365132	Decentralized Zero Trust Network Access	Python/Flask blockchain ledger with SHA-256 chaining and REST API; Ryu-Blockchain Zero Trust middleware with Implicit Deny logic; TensorFlow Lite anomaly detection pipeline with auto-blocking; security evaluation and penetration testing; real-time security dashboard	Python, Flask, Ryu, OpenFlow, Mininet, TensorFlow Lite, scikit-learn
D.W.A.I. Abeynayake	IT22073228	Privacy-Preserving Engagement Monitoring	Three-layer edge-aggregation-cloud architecture design; Laplace	TensorFlow 2.17, DeepFace, OpenCV,

			differential privacy implementation ($\epsilon = 1.3$, IBM diffprivlib); multi-modal engagement detection (DeepFace + OpenCV); 30-second window aggregation with 100% frame deletion; Flask REST API and real-time educator dashboard; performance and privacy validation	diffprivlib, Flask, SQLite, Python
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Table 3.7: Summary of Each Student's Individual Contribution to the Smart Classroom IoT Network Platform

CHAPTER 4: CONCLUSION

4.1 Summary of Achievements

This research has successfully designed, implemented, and evaluated an integrated Smart Classroom IoT Network platform comprising four interdependent components that collectively address the environmental, network management, security, and privacy dimensions of contemporary educational IoT deployment.

The IoT Environmental and Attendance Monitoring component delivered a validated, low-cost hardware-software system operating in a real classroom environment across ten sessions. The multi-sensor ESP32 node achieved a 99% Wi-Fi data transmission success rate and 94.3% RFID attendance accuracy with zero false positives. The Q-Learning middleware achieved 72.8% alert decision accuracy, surpassing the static threshold baseline by 11.4 percentage points and reducing false positive alerts by 8.4 percentage points across 500 evaluated states. The dual MERN-stack application layer maintained sub-second data latency (mean 612 ms) across all tested conditions.

The Intelligent SDN Traffic Classification and QoS component delivered a privacy-preserving, real-time traffic management system capable of differentiating academic from non-academic traffic on encrypted network flows. DSCP-based prioritization and meter-based rate limiting were validated through live traffic testing across academic and non-academic scenarios. The Logistic Regression classifier achieved above 85% accuracy on flow samples with sufficient observation windows, demonstrating practical deployability without the computational overhead or privacy concerns of deep packet inspection approaches.

The Decentralized Zero Trust Network Access component achieved all primary security objectives. The blockchain ledger sustained 847 authorization queries per second at 6.2 ms median response, detected 100% of chain tampering attempts, and maintained authentication operation during simulated cloud outages. Ryu-Blockchain middleware achieved zero false authorizations across 500 authentication tests at 87 ms

median single-device latency. The TensorFlow Lite anomaly detection model achieved a macro-averaged F1-score of 0.922, with automatic DDoS blocking within 12 seconds of attack initiation in 96% of test cases, demonstrating effective post-authentication security complementing the Zero Trust access control layer.

The Privacy-Preserving Engagement Monitoring component achieved all four primary objectives. The $\epsilon = 1.3$ Laplace differential privacy mechanism, 100% frame deletion, and edge-only biometric processing provide a mathematically rigorous and legally defensible privacy posture compliant with GDPR Article 25. The $358,293\times$ bandwidth reduction enables deployment on 2G/3G network infrastructure. The 84% engagement classification accuracy represents only a 6% reduction from the non-private baseline, validating the privacy-utility coexistence hypothesis.

4.2 Limitations

Several limitations constrain the interpretation of the findings and define the boundary conditions of the current implementation.

The single-classroom, single-institution data collection for the IoT environmental monitoring component limits the generalizability of environmental findings and Q-Learning performance to other institutions, building constructions, and geographic locations. The Q-Learning policy has not converged after ten sessions, representing a lower bound of long-term system performance rather than the achieved optimum. The MQ-7 sensor's known humidity cross-sensitivity may have introduced positive CO reading bias in the high-humidity conditions observed throughout monitoring.

The SDN traffic classification system's reliance on observation windows before classification introduces latency for short-lived flows of fewer than 5 packets, which may be misclassified due to insufficient statistical evidence. CDN-based routing and DNS-over-HTTPS reduce contextual feature availability, introducing classification uncertainty for services distributed across shared IP infrastructure. The model requires periodic retraining to maintain accuracy as application traffic patterns evolve.

The DZTNA component's simplified single-node blockchain ledger provides strong tamper-evidence guarantees but does not achieve true Byzantine fault tolerance of a multi-peer Hyperledger Fabric deployment. Physical compromise of the edge gateway device itself remains outside the current threat model. The AI anomaly detection model's training on public benchmark datasets rather than classroom-specific traffic may introduce accuracy discrepancies in deployment environments with significantly different traffic baselines.

The engagement monitoring component's accuracy ceiling is partly constrained by the FER2013-trained DeepFace baseline (70–80% single-frame accuracy), with known dataset limitations including noisy labels and limited demographic diversity. The absence of posture detection — removed due to a MediaPipe-TensorFlow dependency conflict — represents a functional gap relative to the original four-modality design. All performance measurements were conducted on a single development machine rather than a 25–30-student concurrent deployment, limiting validation of multi-client real-world scalability.

4.3 Future Work

Future work is structured across three phases corresponding to the maturation trajectory from validated research prototype to institutionally deployable product.

In the immediate term (months 1–3), the following enhancements are prioritized. Extended Q-Learning training across a full academic semester (70–80 sessions) is required to achieve Q-table convergence across the full 81-state space, with evaluation of Deep Q-Network variants for improved generalization to continuous sensor value spaces. Actuator integration — relay-controlled fan actuation and audio alert in response to Q-Learning Critical decisions — would close the sense-decide-act loop and enable evaluation of physical environmental intervention effectiveness. For the SDN component, integration with additional contextual signals (application-layer protocol identification, time-of-day features) and evaluation in larger network environments would strengthen classification robustness.

In the medium term (months 4–6), the DZTNA security layer should be extended with full Hyperledger Fabric multi-peer deployment to achieve Byzantine fault tolerance, and ECC-based device certificates should replace MAC address authentication to prevent hardware address spoofing by attackers with physical device access. Federated learning for AI model updates across multiple school deployments would enable privacy-preserving continuous improvement of the anomaly detection model. For engagement monitoring, a classroom pilot study involving 25 or more real students is essential to validate multi-client scalability, gather ground-truth engagement correlation data against academic outcomes, and collect cultural adaptation feedback for the emotion-engagement mapping.

In the longer term (months 7–12), the four components should be deployed as a fully integrated platform across three to five partner schools, enabling multi-component correlation analysis (such as environmental CO₂ and temperature correlation with engagement trajectories), institution-wide environmental quality reporting, and commercial viability assessment at realistic deployment scale. Multi-classroom deployment with MongoDB Atlas cloud infrastructure would validate the platform's scalability architecture and provide the evidence base for the commercialization model presented in Chapter 2.

4.4 Concluding Remarks

The Smart Classroom IoT Network platform developed in this research demonstrates that four of the most pressing technical challenges in contemporary educational IoT deployment — environmental quality monitoring, intelligent network management, resilient Zero Trust security, and privacy-preserving engagement analytics — can be addressed simultaneously within a unified, affordable platform suitable for resource-constrained institutions.

The research makes a substantive empirical contribution by validating each component in real or near-real conditions: the IoT monitoring system in a live SLIIT classroom with 30 students and 10 sessions; the DZTNA security system across 500

authentication events and 150 tamper tests; the engagement monitoring system across 2,608 processed frames with Wireshark-validated bandwidth measurement; and the SDN system across live academic and non-academic traffic scenarios. These empirical foundations distinguish this work from purely simulation-based research and provide a credible foundation for the commercialization and deployment arguments advanced throughout.

The broader implication of this research extends beyond smart classrooms. The architectural patterns demonstrated — Q-Learning adaptive IoT middleware, flow-based encrypted traffic classification, Ryu-Blockchain Zero Trust enforcement, and edge differential privacy — are directly applicable to healthcare IoT, industrial automation, and smart city deployments where the same constraints of cloud-independence, privacy protection, limited budgets, and diverse device heterogeneity apply. The research contributes replicable technical blueprints for each of these patterns, validated in an accessible educational deployment context.

As IoT-enabled educational environments continue to expand in scope and complexity, the security of the networks connecting these devices, and the privacy of the student behavioural data they generate, become increasingly critical societal concerns. This research demonstrates that it is possible to build smart classroom infrastructure that is simultaneously intelligent, secure, privacy-respecting, and accessible to the institutions that need it most — providing a technically rigorous and practically deployable response to one of the most significant challenges at the intersection of education and technology.

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APPENDIX A: DZTNA SYSTEM ARCHITECTURE DIAGRAMS

Figure A.1 illustrates the complete three-layer DZTNA system architecture. The Identity Layer (bottom) depicts the Python/Flask blockchain ledger with SHA-256 linked block chain, REST API endpoints (/register, /revoke, /check), and SQLite persistent storage. The Enforcement Layer (middle) shows the Ryu SDN controller connected to the Open vSwitch data plane via the OpenFlow 1.3 control channel, with the Ryu-Blockchain middleware component intercepting Packet-In events and performing REST API queries to the blockchain layer before installing or withholding flow rules. The Intelligence Layer (top) shows the TensorFlow Lite anomaly detection process monitoring the OVS mirrored traffic port, the feature extraction pipeline, the trained model, and the alert feedback channel to the Ryu controller.

Figure A.2 presents the Ryu-Blockchain Packet-In processing flowchart. The process begins with the OVS switch receiving a packet from a classroom device and generating a Packet-In event. The Ryu event handler extracts the eth.src MAC address and issues a GET /check request to the blockchain REST API. If authorized: True, Ryu installs a bidirectional flow rule (idle_timeout = 30 s) and forwards the packet. If authorized: False, no action is taken, and the packet is silently dropped — illustrating the Zero Trust Implicit Deny boundary.

Figure A.3 shows the complete network communication sequence diagram for the device authentication and revocation lifecycle: device power-on; first packet and Packet-In event; blockchain query and authorization response; flow rule installation; active session with 30-second re-verification cycles; administrator revocation command; blockchain revocation block append; next re-verification returning authorized: False; flow rule expiration; and subsequent packets being dropped.

APPENDIX B: IOT SENSOR NODE WIRING SCHEMATIC

Figure B.1 illustrates the complete wiring schematic for the ESP32 IoT sensor node. The schematic shows the ESP32-WROOM-32 development board as the central hub, with the following connections: DHT22 temperature-humidity sensor connected to GPIO4 with a 10 k Ω pull-up resistor to 3.3V; MQ-7 CO sensor module analog output connected to GPIO34 with voltage divider conditioning; MFRC522 RFID module connected via SPI (SCK $\rightarrow$ GPIO18, MOSI $\rightarrow$ GPIO23, MISO $\rightarrow$ GPIO19, SDA/SS $\rightarrow$ GPIO5, RST $\rightarrow$ GPIO22, 3.3V power); and analog sound sensor module connected to GPIO35. All sensor modules share the 3.3V power rail and common ground with the ESP32 board. The breadboard prototype was constructed according to this schematic prior to formal testing.

Table B.1 presents the complete bill of materials for a single IoT sensor node deployment.

Component	Approx. Cost (LKR)	Notes
ESP32-WROOM-32 Development Board	1,400	Includes USB-to-UART bridge
DHT22 Temperature & Humidity Sensor Module	350	Includes 10 k Ω pull-up resistor
MQ-7 CO Sensor Module	450	On-board ADC and heater circuit
MFRC522 RFID Kit (1 reader + 5 MIFARE tags)	700	Additional tags ~80 LKR each
Analog Sound Sensor Module (LM393)	250	Dual digital/analog output
Breadboard + Jumper Wires	350	Prototyping; PCB fabrication ~2,000 LKR
ABS Plastic Enclosure Box	400	Standard project box
Total per node	~3,900	~USD 13 at current exchange rate

Table B.1: Hardware Component Bill of Materials — Single IoT Sensor Node

For institutional-scale deployment across 20 classrooms, total hardware cost is approximately LKR 78,000 (USD 260), compared to USD 4,000–20,000 for equivalent commercial environmental monitoring solutions.

APPENDIX C: ENVIRONMENTAL DASHBOARD SCREENSHOTS

Figure C.1 depicts the Environmental Dashboard real-time status panel interface. The panel displays four sensor readings with colour-coded circular indicators: Temperature (green for $\leq 25^{\circ}\text{C}$, amber for $25\text{--}30^{\circ}\text{C}$, red for $>30^{\circ}\text{C}$), Humidity (green for $\leq 60\%$ RH, amber for $60\text{--}70\%$, red for $>70\%$), CO Level (green for ≤ 15 ppm, amber for $15\text{--}35$ ppm, red for >35 ppm), and Noise Level (green for ≤ 50 dB, amber for $50\text{--}65$ dB, red for >65 dB). The most recent RFID attendance event is shown at the bottom of the panel. The 5-second polling interval ensures near-real-time environmental awareness.

Figure C.2 presents the time-series historical data chart rendering, showing a 60-minute rolling window for all four environmental parameters on a single composite Chart.js visualization with dual Y-axes (temperature and humidity on the left, CO ppm and noise dB on the right). The Q-Learning agent's alert decisions are overlaid as annotated vertical markers, distinguishing Advisory alerts (amber) from Critical alerts (red) and enabling visual correlation between environmental parameter trajectories and alert decisions.

APPENDIX D: ENGAGEMENT MONITORING JSON OUTPUT

SAMPLE

The following listing presents the structure of a typical anonymised engagement summary JSON payload transmitted from the student edge client to the educator dashboard server, following application of the Laplace differential privacy mechanism. The payload demonstrates the complete data that crosses the network boundary — no facial imagery, emotion detection intermediates, or biometric data of any kind is included.

```
{ "session_id": "SESSION_001", "window_seq": 3, "timestamp": 1714012860,
  "engagement_proportions": { "ENGAGED": 0.6142, "NEUTRAL": 0.2318,
    "DISENGAGED": 0.1540 }, "visibility_score": 0.8876, "movement_score": 0.7234,
  "composite_score": 71.4, "engagement_tier": "ENGAGED", "epsilon_consumed": 1.3,
  "frames_deleted": 720 }
```

The payload size of approximately 2,668 bytes per 30-second window, transmitted as a single HTTP POST request, produces the 0.20 Kbps per-student bandwidth consumption measured in Chapter 3. The frames_deleted field, confirmed at 720 for each complete window, provides an audit record of the 100% biometric data elimination commitment. The epsilon_consumed field enables cumulative privacy budget accounting across sessions, supporting future implementations of adaptive privacy budget management across extended deployment periods.